10-21 第2篇

阅读理解（说明文） Why Do Cats Almost Always Land on Their Feet?

① If you have ever lived with a cat, you have probably seen a small, quiet magic trick: your cat walks along the top of a sofa, misjudges a step, falls in a flash — and still lands **neatly on all four feet**, as if nothing happened. Scientists call this skill the "**righting reflex**". It is not magic. It is **biology and physics working together in less than a second.**

② The story begins with the **inner ear**. Deep inside a cat's head, three tiny curved tubes are filled with fluid. When the cat tilts or falls, the fluid moves and tells the brain **which way is up**. This sense works so fast that a cat needs only about **0.05 seconds** to know her body is turning the wrong way in the air — faster than you can blink.

③ After the brain says "wrong way up", the cat's **flexible body** begins a surprising dance. She does not have a rigid spine like a rabbit or a dog; she has a **very bendy backbone** with no fixed collarbone. She splits her body into two halves. First, she **tucks in her front legs and stretches out her back legs**, and twists her upper half so that her head faces the floor. Then, in a second move, she **stretches out her front legs and tucks in her back legs**, and twists her lower half to match. From the outside it looks like one smooth roll, but in fact it is two clever little twists, hidden in a single heartbeat.

④ What happens next makes use of **air**. The cat spreads her **legs** and **fluffy tail** to the side, like a tiny parachute. This slows her fall and gives her more time to finish the turn. She also bends her knees and elbows **like four small springs**, so that when she lands, the shock is **spread across her paws, wrists, shoulders and spine**, not sent straight into her head.

⑤ Is there a rule about **how high** a cat can fall safely? Vets call it "**high-rise syndrome**" (高楼综合征). Surprisingly, most cats who fall from **between 3 and 6 floors** are badly hurt — because the cat is terrified and has no time to fully twist and relax. However, cats that fall from **over 7 floors** are often *less* badly hurt than you might think. From that height, they reach a steady speed (called **terminal velocity**), relax, open their legs wide like a flying squirrel, and land with their weight spread. In New York, vets once studied 132 cats that had fallen from tall buildings; the ones from the **highest floors** actually had **fewer broken bones** than those from the middle floors.

⑥ Still, this **does not** mean cats are safe to fall. Even a successful landing may crack ribs or hurt the jaw. Many vets remind us of a rule: **windows on the 2nd floor and above should always have safety nets** if a cat lives inside. A cat can do clever mid-air physics, but she cannot choose the floor, the weather, or the road under the window.

⑦ The cat's righting reflex has taught humans more than we can count. Astronauts on the International Space Station once watched slow-motion films of cats falling and learned about **mid-air rotation** with no outside push — a question that had puzzled physics classrooms since the 1890s. Today, rescue robots and even some satellites copy the idea of "two halves turning in opposite directions" to right themselves without using fuel.

⑧ So next time you see your cat jump awkwardly from the back of a sofa and land softly on four feet, remember: you are watching a **two-hundred-year-old physics problem** being solved, silently, by a creature that only wants her breakfast.

注释： righting reflex 翻正反射；inner ear 内耳；flexible 灵活的；terminal velocity 终端速度；high-rise syndrome 高楼综合征

根据短文内容，选择最佳答案。

1. Which body part first tells a falling cat which way is up?

A. Its tail.
B. Its eyes.
C. The fluid-filled curved tubes in its inner ear.
D. Its claws.

2. How does the cat's flexible body help during the fall, according to Paragraph ③?

A. It allows her to grow wings.
B. It allows her to twist her upper half first, then her lower half, turning her body in the air through two clever little moves.
C. It allows her to stop falling.
D. It allows her to become smaller.

3. What does the cat use as a "tiny parachute"?

A. Her fur colour.
B. Her spread legs and fluffy tail.
C. Her whiskers.
D. Her claws only.

4. According to the "high-rise syndrome" study, which group of cats were often less badly hurt than those in the middle?

A. Cats from the 1st floor.
B. Cats from the 3rd to 6th floors.
C. Cats from over the 7th floor, who could relax at terminal velocity and spread out their

weight.
D. Cats who did not fall at all.

5. What is the main message of the whole passage?

A. Cats should be sent to space.
B. The cat's righting reflex is a piece of clever biology and physics; it has taught humans about mid-air rotation and now helps robots, though cats still need safety nets on windows.
C. Cats never get hurt when they fall.
D. Cats should not live in tall buildings at all.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段 "three tiny curved tubes are filled with fluid ... tells the brain which way is up"
【解析】内耳首先告知方向。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "twists her upper half ... then ... twists her lower half"
【解析】两段式分体翻转。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "spreads her legs and fluffy tail to the side, like a tiny parachute"
【解析】直引。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑤段 "cats that fall from over 7 floors ... fewer broken bones"
【解析】高楼反而伤较轻。A、B、D 均错。故选 C。

5.
【答案】B
【题型】主旨大意题
【定位】第⑦⑧段总结 + 第⑥段安全提醒
【解析】综合概括科学价值与现实提醒。A、C、D 均夸张或片面。故选 B。